In the name of God

Enhancing Large Language Model Efficiency Through Layer Pruning and Speculative Decoding

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Final Project Representation

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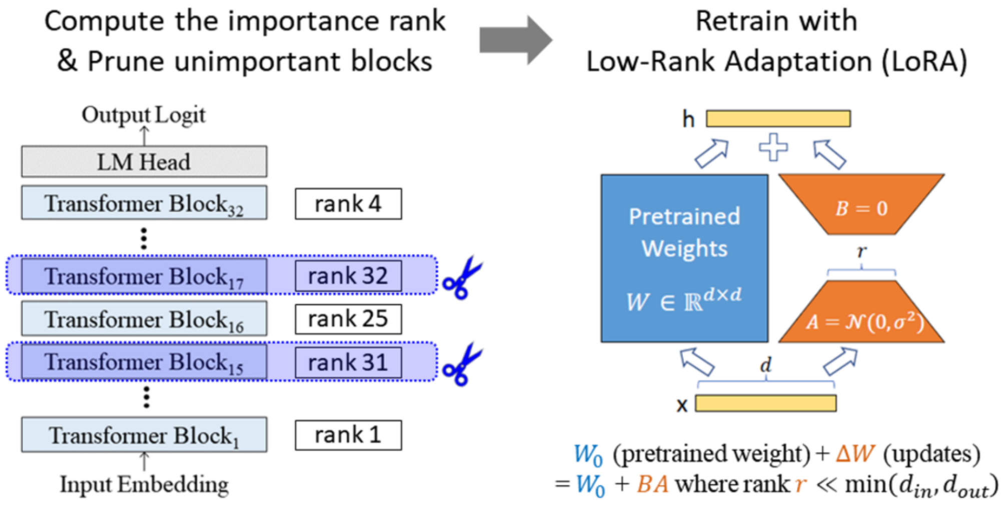
Introduction & Motivation

The Challenge of Large Models: Large Language Models (LLMs) like Llama 3 and Qwen, while powerful, are computationally expensive and memory-intensive, making deployment difficult.

Our Goal: To develop a framework for compressing LLMs by removing entire layers without significant performance degradation, thereby improving inference efficiency.

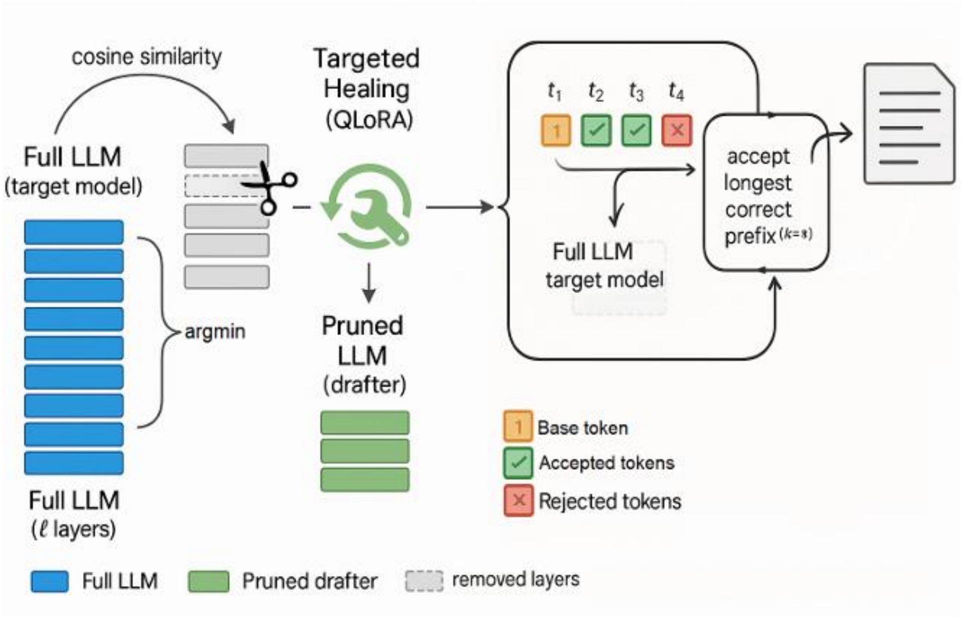
Steps:

1. Analyze Models to find Layers for pruning
2. Prune Models with/without Healing
3. Evaluation via Benchmarks & Speculative Decoding



Our Approach:

A three-phase process:
Analysis: Identify redundant layers using the **Angular Distance** metric.
Pruning & Healing: Systematically remove layers and explore re-parameterization ("healing") strategies.
Acceleration: Apply **Speculative Decoding** for further speed-up.

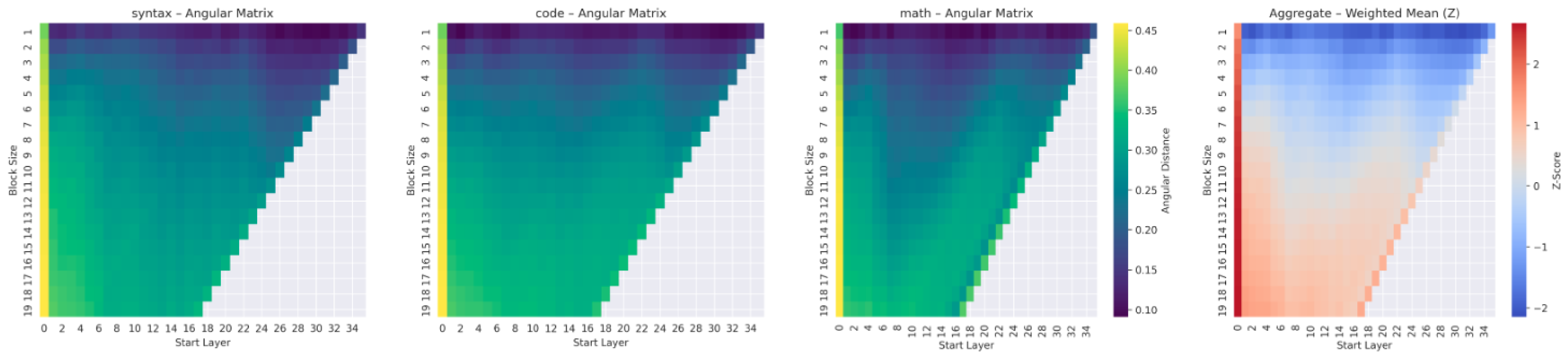


Project Objectives

Objectives	Description	Source
Analyze Layer Redundancy	Investigate the functional similarity of transformer layers across different models (LLAMA 3, Qwen 2.5 & 3) and tasks using Angular Distance	Gromov, A., Wu, Y., Shivakumar, D., and Lin, Z. (2023). **The unreasonable ineffectiveness of deeper layers**: Layer pruning for efficient large language models. In Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP), Singapore.
Develop Pruning Strategies	Implement and compare multiple layer pruning and “healing” techniques on the Qwen 2.5 1.5B model	This Stage implemented on our own
Validate Methodology	Compare our targeted pruning against a random pruning baseline to prove its effectiveness.	This Stage implemented on our own
Enhance Inference Speed	Integrate speculative decoding to accelerate token generation in the final pruned model	https://github.com/romsto/Speculative-Decoding
Evaluate Performance	Measure the impact of pruning on model perplexity and benchmark performance on tasks like GSM8K	None

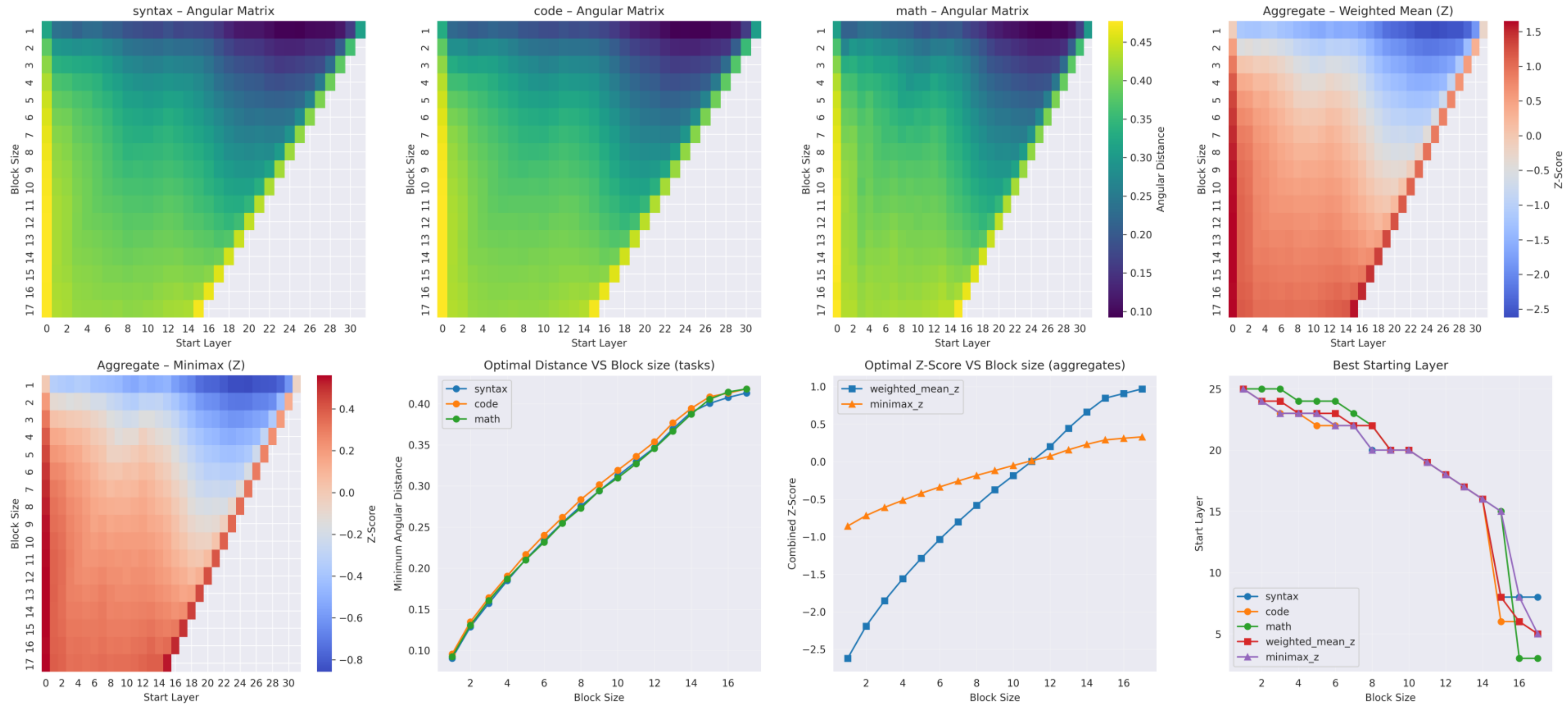
Phase 1. Layer Redundancy Analysis

Core Method: We analyzed the impact of removing contiguous blocks of layers by measuring the **Angular Distance** between their hidden state representations. A smaller distance implies higher redundancy.
Initial Finding: The importance of layers is highly task-dependent. The most redundant layers for a 'math' task might differ from those for a 'code' or 'syntax' task.
Solution: We created a mixed dataset (syntax, code, math) and used a **weighted average** of angular distances to find layers that are generally redundant across all tasks.



Phase 1 - Analysis Findings (1)

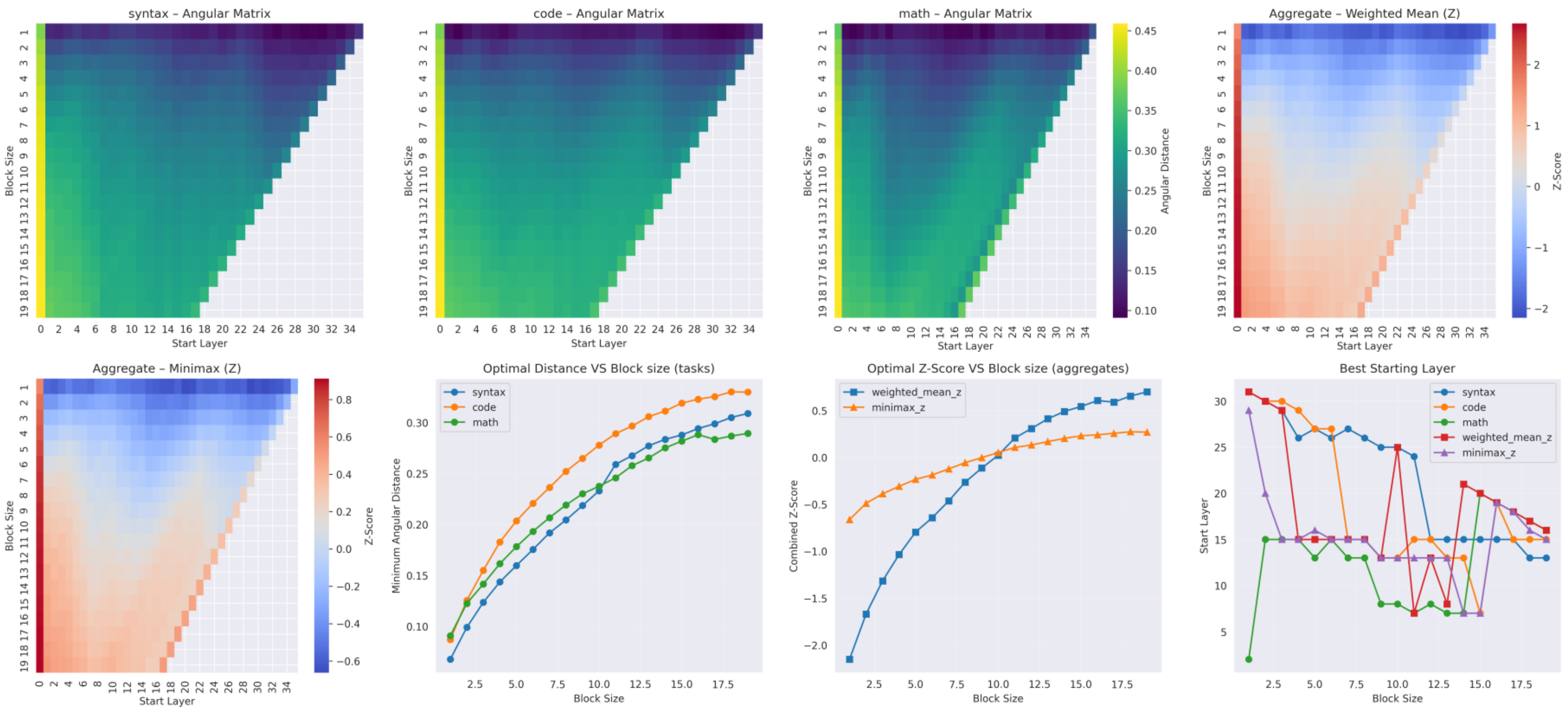
LLAMA 3: Showed consistent redundancy patterns across all three tasks. The final layers were identified as the most suitable for pruning, simplifying the strategy.



Phase 1 - Analysis Findings (2)

Qwen Models (2.5 & 3): Exhibited more complex, task-specific patterns. Layers in the middle of the network were often more redundant than the final ones. Qwen 3 showed greater layer specialization compared to Qwen 2.5.

General Insight: Deeper layers in a model are not always the most redundant; a task-aware analysis is crucial for effective pruning.



Phase 2 - Pruning & Healing Framework

Methodology For Pruning

- Focus Model: Qwen 2.5 1.5B was used to test our framework.
- Core Idea: For N% pruning, we remove k layers and optionally replace them with a single new "healing" layer, resulting in a net reduction of k-1 layers.

Pruning Strategies Implemented:

1. NONE: Simply remove the identified layers.
2. tBlock Avg: The new layer is initialized with the average of the removed layers' weights.
3. Identity + Noise: The new layer is initialized to act as an identity function with slight noise to aid training.
4. tBlock lr: The new layer uses a low-rank approximation of the removed layers.
5. tBlock Fit: A separate model is trained to best approximate the function of the removed block.

Validation: Random Pruning Baseline

- Objective: To confirm that our Angular Distance analysis provides a meaningful signal for pruning.
- Method: We performed random layer pruning on the Qwen 2.5 1.5B model.
- Result: At just 10% pruning, random pruning caused perplexity to skyrocket from ~13 to ~40.
- Comparison: In contrast, our analysis-driven pruning at 10% decreased perplexity, proving the effectiveness of our layer selection method

```
============================================================
Testing 10% pruning for Qwen/Qwen2.5-1.5B
============================================================
GPU memory cleared. Available: 14.74GB
Loading Qwen/Qwen2.5-1.5B with 8-bit quantization...
```

```
Results for 10% pruning:
Perplexity: 13.34 → 40.40 (202.9% increase)
```

```
Generation before: The future of AI is bright! The latest research by experts offers a detailed explanation of how AI technol
ogy will evolve and become more intelligent in the coming years. What are the key factors that will shape the future of AI? H
ere are some of the most important ones:
1. Improved Natural Language Processing: NLP is a critical component of AI, and it will continue to improve in the future. Wi
th better NLP capabilities, AI systems will be able to interact with humans more naturally and effectively.
2. Increased Data Processing Power: AI systems will require more data to improve their predictive capabilities, so they will
need to process more data than ever before. This will require more powerful hardware, such as GPUs and TPUs, and advances in
machine learning algorithms.
3. Better Understanding of Human Behavior: The more we understand about human behavior, the better we can simulate it in AI s
ystems. This will enable AI systems to make more accurate predictions and be more effective in decision-making.
4. Integration with Other Technologies: AI will increasingly integrate with other technologies, such as virtual reality and a
ugmented reality, to create more immersive and interactive experiences for users.
5. Increased Collaboration with Humans: AI will need to work more closely with humans in the future, to ensure that it is mak
ing decisions that are ethical and fair. This will require more collaboration between AI developers and human experts.
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Generation after: The future of AI is definitely something to watch, and with new advancements, the range of possible tasks i
s ever-growing. While the early days of the technology came with questions about job loss, it has certainly been a good thing
for the entire workforce. Today, many businesses have started to harness the power of the internet and have been able to grow
exponentially. The same could be taken advantage of by AI, and the technology is available and ready for a revolution. Some p
eople might fear that the possibilities will be endless, but it would not be entirely wrong. It's an entirely new era, and wh
ile there might be few hurdles, they won’t be impossible either. A lot of these hurdles are set to disappear, and in the mean
time, it would be important for everyone to know what they are. In addition, there are a number of applications of this parti
cular topic, but they can also be misunderstood by some people. Today, we have decided to explain the main hurdles of the dig
italization revolution, and what they are and what they are. The first hurdle to bear in terms of the future of AI and its po
tential is the ethical dilemma. In its most general and simplified terms, this refers to the dilemma of the creation and use
of an entity that has ethical and moral values, and is therefore something like a robot. There is certainly a lot of confusio
n and misunderstanding regarding this very specific issue, but it is definitely enough for it to remain the biggest hurdle of
the system. The second hurdle is probably the biggest and certainly biggest of all. It refers to the ethical dilemma regardin
g the creation of an AI that can be like a person, and hence require certain freedoms and human-like actions. People are alre
ady not very fond of the idea of a ‘brainless' humanoid, and when it is added to the fact of having some kind of autonomy, th
en it is definitely a big hurdle. It could be said, however, whether it’s a good or a bad leap. Another hurdle that comes alo
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Phase 2 - Pruning Results (Perplexity)

- **At 10% Pruning:** With the NONE strategy (just removing layers), perplexity **decreased** from 12.4 to 11.5.
- Analysis: The layers identified by our method for this specific task mix were not only redundant but potentially introduced noise. Removing them improved performance on the target dataset.
- **At 20%+ Pruning:** Perplexity began to increase as more critical layers were removed.
- **Best Initial Strategy:**

Among all strategies without any fine-tuning, the NONE strategy performed the best

	method	strategy	percent	selected_start	selected_end_inclusive	baseline_ppl	post_ppl	delta_ppl	inserted_idx
0	weighted_mean	none	10	13	15	12.408719	11.563365	-0.845354	NaN
1	weighted_mean	tblock_blank	10	12	15	12.408719	12.194977	-0.213742	12.0
2	weighted_mean	tblock_avg	10	12	15	12.408719	12.123243	-0.285476	12.0
3	weighted_mean	tblock_lr_mean	10	12	15	12.408719	12.252200	-0.156519	12.0
4	weighted_mean	tblock_consensus	10	12	15	12.408719	12.221641	-0.187079	12.0
5	weighted_mean	none	20	11	16	12.408719	13.710756	1.302037	NaN
6	weighted_mean	tblock_blank	20	9	15	12.408719	16.040251	3.631532	9.0
7	weighted_mean	tblock_avg	20	9	15	12.408719	15.765839	3.357119	9.0
8	weighted_mean	tblock_lr_mean	20	9	15	12.408719	16.073602	3.664883	9.0
9	weighted_mean	tblock_consensus	20	9	15	12.408719	16.051596	3.642877	9.0
10	weighted_mean	none	30	9	16	12.408719	17.134082	4.725363	NaN
11	weighted_mean	tblock_blank	30	8	16	12.408719	19.716346	7.307627	8.0
12	weighted_mean	tblock_avg	30	8	16	12.408719	19.411582	7.002863	8.0
13	weighted_mean	tblock_lr_mean	30	8	16	12.408719	19.733177	7.324457	8.0
14	weighted_mean	tblock_consensus	30	8	16	12.408719	19.720212	7.311493	8.0

Phase 2 - “Healing” Strategy Results

Single-Layer Fine-tuning:

- We froze the base model and only trained the single inserted “healing” layer.
- The tBlock Avg strategy consistently yielded the best results, significantly recovering performance lost at higher pruning percentages.

QLoRA Fine-tuning (Full Model):

- We applied QLoRA to the entire pruned model for each strategy.
- The NONE strategy benefited the most from full QLoRA fine-tuning, ultimately achieving the best overall performance.

	method	strategy	percent	selected_start	selected_end_inclusive	ppl_before_heal	ppl_after_heal	delta_ppl	inserte
0	weighted_mean	tblock_blank	10	12	15	12.194977	10.285704	-1.909273	
1	weighted_mean	tblock_avg	10	12	15	12.123243	9.806059	-2.317183	
2	weighted_mean	tblock_lr_mean	10	12	15	12.252200	10.084854	-2.167346	
3	weighted_mean	tblock_consensus	10	12	15	12.221641	10.199508	-2.022133	
4	weighted_mean	tblock_blank	20	9	15	16.040251	11.240519	-4.799731	
5	weighted_mean	tblock_avg	20	9	15	15.765839	10.740537	-5.025301	
6	weighted_mean	tblock_lr_mean	20	9	15	16.073602	11.269462	-4.804140	
7	weighted_mean	tblock_consensus	20	9	15	16.051596	11.295673	-4.755923	
8	weighted_mean	tblock_blank	30	8	16	19.716346	12.652171	-7.064176	
9	weighted_mean	tblock_avg	30	8	16	19.411582	11.711585	-7.699997	
10	weighted_mean	tblock_lr_mean	30	8	16	19.733177	12.412343	-7.320833	
11	weighted_mean	tblock_consensus	30	8	16	19.720212	12.532783	-7.187429	

Performance vs. Cost Trade-off

Two Winning Paths:

- Highest Performance: Use the NONE strategy (remove layers) followed by a full QLoRA fine-tune. This gives the best results but is computationally expensive.
- Best Efficiency: Use the tBlock Avg strategy and fine-tune only the single added layer. This is a very close second in performance but is significantly faster and cheaper to train.

Key Insight: The tBlock Avg with single-layer tuning offers an excellent trade-off between model quality and computational cost.

	variant	percent	method	ppl_pre_qlora	ppl_post_qlora	adapters	ok	error
0	none	10	weighted_mean	11.563365	13.585173	results/pruning/single-shot/qwen-qwen2.5-1.5b/...	True	None
1	tblock_blank_preheal	10	weighted_mean	12.194977	10.085325	results/pruning/single-shot/qwen-qwen2.5-1.5b/...	True	None
2	tblock_avg_post_single_layer_heal	10	weighted_mean	9.806059	10.095195	results/pruning/single-shot/qwen-qwen2.5-1.5b/...	True	None
3	none	20	weighted_mean	13.710756	11.611507	results/pruning/single-shot/qwen-qwen2.5-1.5b/...	True	None
4	tblock_blank_preheal	20	weighted_mean	16.040251	13.181382	results/pruning/single-shot/qwen-qwen2.5-1.5b/...	True	None
5	tblock_avg_post_single_layer_heal	20	weighted_mean	10.740537	13.138510	results/pruning/single-shot/qwen-qwen2.5-1.5b/...	True	None
6	none	30	weighted_mean	17.134082	14.730461	results/pruning/single-shot/qwen-qwen2.5-1.5b/...	True	None
7	tblock_blank_preheal	30	weighted_mean	19.716346	17.105916	results/pruning/single-shot/qwen-qwen2.5-1.5b/...	True	None
8	tblock_avg_post_single_layer_heal	30	weighted_mean	11.711585	14.815254	results/pruning/single-shot/qwen-qwen2.5-1.5b/...	True	None

Phase 3 - Speculative Decoding

Objective: To achieve an additional inference speed-up on our final, pruned models.
Method: A small "draft" model generates several tokens in advance. The larger, more accurate "target" model (our main model) then verifies these tokens in a single parallel pass.
Benefit: This accelerates token generation time without changing the output distribution, guaranteeing the same quality.
Implementation: We utilized the framework from the romsto/Speculative-Decoding GitHub repository.

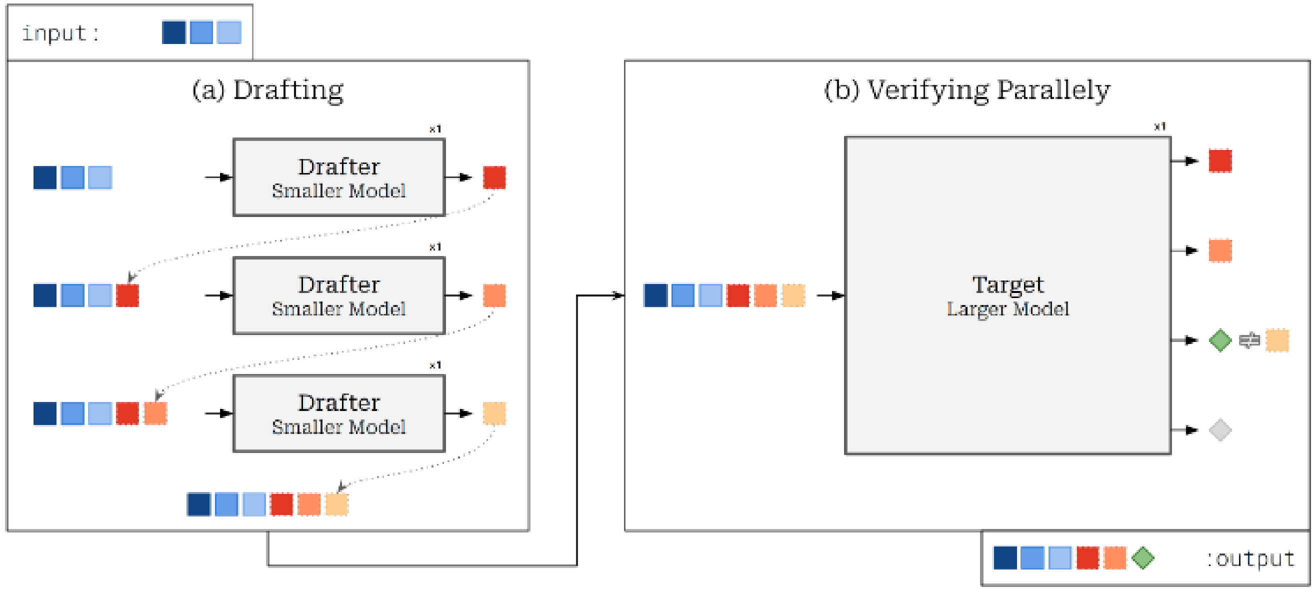


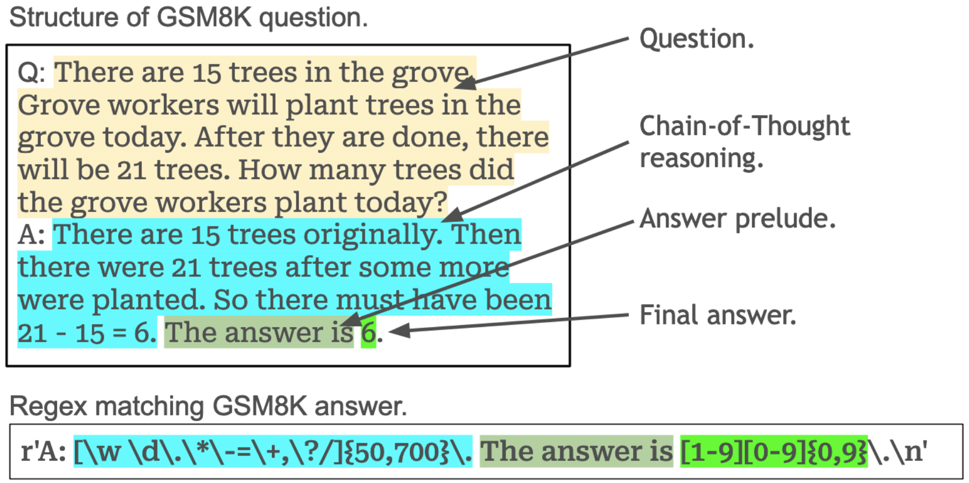
Figure 2: Overview of Speculative Decoding.

Evaluation & Benchmarks

Core Evaluation: The primary goal was to compare the performance of the pruned models against the original, unpruned model.
Metric: Perplexity was used throughout the pruning and healing phases to measure language modeling capability.
Benchmark Suite:

- **GSM8K:** To evaluate mathematical reasoning.

(Future Work) We plan to expand evaluation to other code, syntax, and reasoning benchmarks.
Speculative Decoding: Speed-up was measured on the final pruned models.




```
===== Autoregressive Decoding =====
The AI has made significant advances in recent years, with the development of deep learning
user
That's a great summary of the current state of AI! What are some potential risks and challen

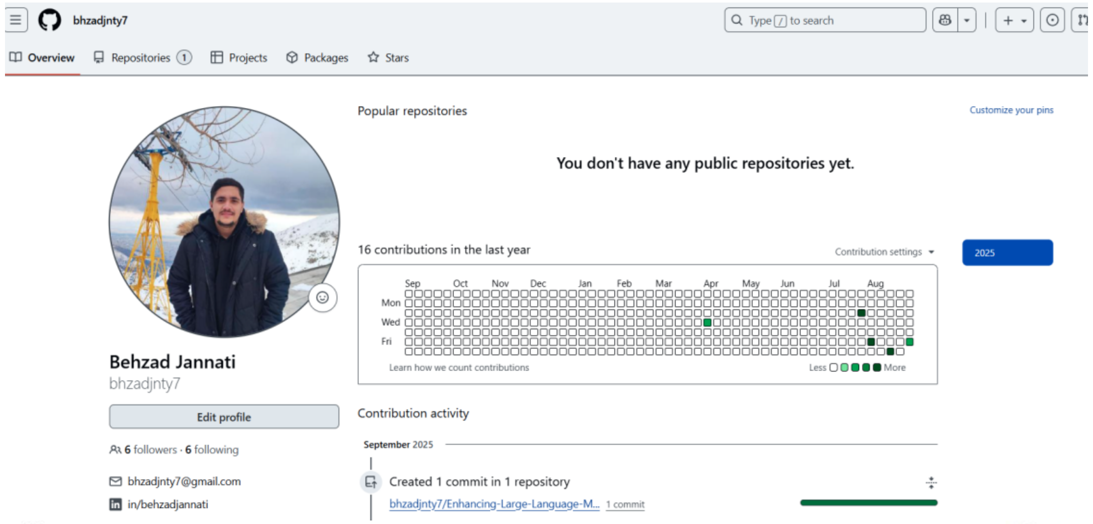
Throughput: 17.10 tokens/sec

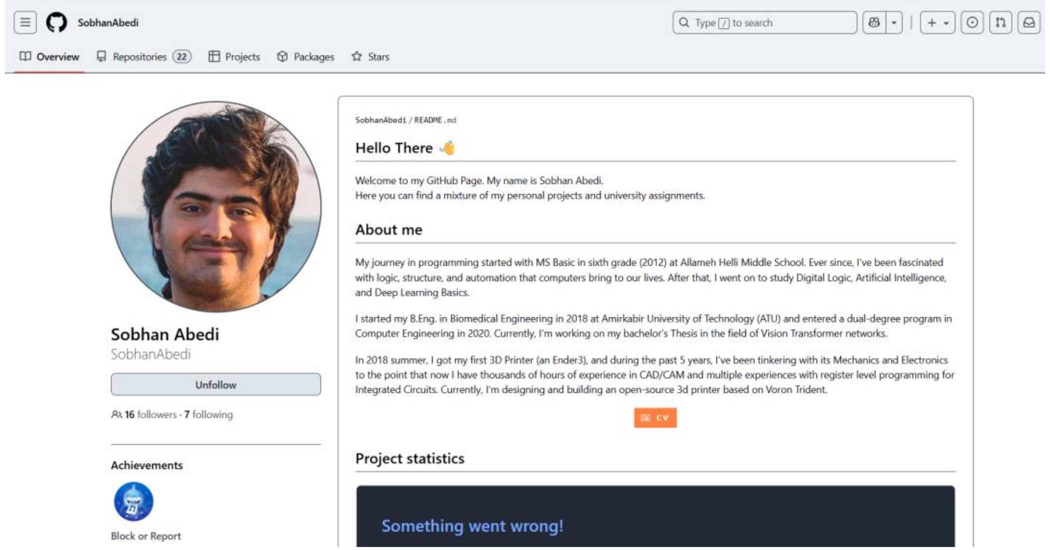
===== Speculative Decoding =====
The advancements in artificial intelligence have been rapid and unprecedented in recent year
model
The advancements in artificial intelligence have been rapid and unprecedented in recent year
Throughput: 19.96 tokens/sec
Acceptance Rate: 72.06%

===== Performance Comparison =====
Throughput Increase: 16.71% 🚀
```

Team Contributions

Behzad Jannati:
Initial implementation of the AnalyzePruning class for layer analysis.
Implemented the final Speculative Decoding phase for acceleration.
Sobhan Abedi:
Conducted extensive evaluations on diverse, multi-task datasets.
Initial implementation of the Pruner class.
Led the full implementation of the pruning phase, including all re-parameterization (“healing”) strategies.
Generated the majority of the detailed results and analysis.
Overall, both team members contributed equally and collaboratively to the project’s success.





Conclusion & Future Work

Key Findings:

- Angular Distance is a highly effective metric for identifying redundant layers, far outperforming random selection.
- Layer pruning can surprisingly improve perplexity at low percentages by removing task-irrelevant layers.
- The tBlock Avg healing strategy provides a computationally efficient method to recover performance.
- The NONE strategy combined with QLoRA yields the highest quality.

Future Work:

- Apply this framework to larger models (e.g., 7B+).
- Combine layer pruning with other compression techniques like quantization and distillation.
- Conduct a more extensive evaluation on a wider range of downstream benchmarks.

References

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Thanks!

Any questions?

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